

BHASKAR GARG

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FIELDS OF INTERESTS

VLSI, Physical Design, Embedded Systems

EDUCATION

IIT, ROORKEE

Bachelor of Technology,
Department of Electrical Engineering.

July 2018 - May 2022

CGPA: 8.1

JOB EXPERIENCE

SoC Design Engineer, Intel Technologies

June 2022-Present

- **TFM Owner for Critical IP with more than 100 partitions ensuring quality execution adhering to tight schedules**

[noitemsep, leftmargin=-1]Creating the Flows and Methodologies to be consumed by the Physical Design Engineers and supporting them. Enabling a mixed flow containing advanced tools like Fusion Compiler, Genus and Innovus. Working with the RTL Team to ensure elab clean and mapped netlist to minimize back and forth corrective communications for timely IP delivery. Working as per the self-curated TFM plan to evolve TFM by different stages of projects involving collaboration with various Domain owners, Tool Teams, AEs and Engineers.

- **Implementing LLM based systems and workflows to Leverage AI and target for Engineering Execution Efficiency Improvement**

[noitemsep, leftmargin=-1]Implementing **LLMs**(Large Language Models) in the BackEnd Methodologies, Experimenting with various possible use cases like **auto analysis of design reports** on various stages of Flow. Created a Novel Multi-Agentic Workflow to minimize hallucinations or inaccuracies of LLM results. Filed an IDF(Invention Disclosure Form) for the same, But Intel chose to keep it as Trade Secret instead of making it publicly available through Patent. Awarded with Distinguished Invention Award for the same.

- **Created an automated app to port memory views across tech nodes to enable PnR work in early stage of projects**

[noitemsep, leftmargin=-1]Understood the gap of unavailability of memories in the advanced tech node. Levered TCL scripting and LM(Library Manager) shell to convert the libraries and generate the compatible NDMs. Saved Months of Bandwidth for the entire SD team to early project Tech Readiness activities with the available RTL and ported Memories. Target node was **intel foundry** and parent project was in **TSMC node**.

- **Proficient in building end to end flows for the Design Teams**

[noitemsep, leftmargin=-1]Engineered an **automated VLSI design flow**, leveraging RTL to GDSII processes, and significantly enhanced chip design efficiency using Synopsys Fusion Compiler and TCL scripting, increasing **design process efficiency to 5x**. Proficient in using Fusion Compiler and Innovus for backend VLSI design, enhancing design quality and efficiency.

INTERNSHIPS

- **MITACS Research Scholar @ University Of Windsor, Canada** June 2021-August 2021

· Designed 4 different Micromachined Gas sensors in COMSOL MultiPhysics for electronic nose.

- Simulated the designs to record their change in Capacitance and resistance in different environments.
- Studied Sensor design and simulations processes in COMSOL MultiPhysics for sensor instrumentation

Innovation Engineer, HVM Drives

Jan 2021 - June 2021

- Developed an IOT based device to communicate Induction Motor parameters to users using WhatsApp.
- The motor switches(using Relays) through Whatsapp commands only(via Twilio and MQTT protocol).
- Whatsapp based IOT device could give a feel of true virtual assistant present at the Motor station.
- Indian Patent filed and granted for the developed communication system.

PROJECTS

Technical Stack for Industrial Sensor reading—BHEL(Bharat Heavy Electrical Limited) March 2021-March 2022

- Designed an Atmega2560 based PCB to interpret a total of 24 Industrial Sensors input.
- The PCB featured Communication of sensor data to the PC or the next PCB.
- PC read the inputs and controlled the MUXs using a User-friendly GUI based on Firmata protocol.

IOT based health monitoring system—DST(Department of Science and Technology, GOI) March 2021-May 2021

- Built an IOT based system to monitor essential biological parameters(Respiration, Pulse, SPO2, Temp).
- ESP8266 was used to communicate the input data from sensors(AHT10, MAX30100, Flex Sensor).
- Self-built Data receiver on lab server to display the received data on the Website using a static IP.

Air Quality Monitoring System Using FPGA SPARTAN-3E—Electrical Engr. Dept, IITR September 2020-January 2021

- Generated a 32.768KHz frequency that would vary because of the PM2.5 particles sticking on the MQTF(Micro Quartz Tuning Fork).
- Used Onboard 5MHz oscillator or FPGA to measure this change in MQTF frequency hence the PM2.5.

Real-Time Posture Correction—CSE Dept, IITR

September 2019-May 2020

- Developed a wearable-affordable device to prevent the rising problem of back pain among youngsters.
- Also developed software to detect the wrong posture using a laptop camera while working.

INNOVATION AND INVENTIONS

Patent Granted for developing "System and Method for monitoring and Controlling a Plurality of Field Devices"

Distinguished Invention Award for the work titled as "Multi-agent workflow: a new era of chip design"

Copyright granted for "Software System for Real-Time Management of Sitting Posture", in Copyright Office Govt. of India, Department for promotion of Industry Internal Trade, Govt of India, 2021

TECHNICAL STRENGTHS

Computer Languages

C++, Python, TCL

Software & Tools

Fusion Compiler, Innovus, Genus, LT Spice, Xilinx, Altium, COMSOL, MATLAB

Skills

Proficiency in Basic PnR flows like Placement, CTS, Routing debug and analysis.

ACADEMIC ACHIEVEMENTS

Winner at IIT Roorkee Hult Prize 2020 organised by Bill Clinton Foundation.

Chanakya UG Fellowship by iHUB DivyaSampark 2021

P2P(Prototype to Product) award by Design Innovation Centre, IIT Roorkee 2020

Gold Medal in Intel IIT Tech Meet 2019.

All India Rank 8 in National Creative Olympiad 2015-16

Gold medal in State and District level of National Children Science Congress 2014.

Haryana's State Child Scientist Award 2013.

EXTRA-CIRRICULAR

National level Chess player in 2014-16.

Volunteer at Sanskrit Club, IIT Roorkee.

Vice President of Electrical Engineering Student Society.

Member at IEEE Student Chapter, IIT Roorkee